A. Appendix

A. Templates

Addition	**Subtraction**
{A} + {B} = {C}	{A} - {B} = {C}
{A} plus {B} equals to {C}	{A} minus {B} equals to {C}
The addition of {A} and {B} is {C}	The difference of {A} and {B} is {C}
The addition of {A} and {B} equals to {C}	The difference of {A} and {B} equals to {C}
The addition of {A} and {B} equals to {C}	The difference of {A} and {B} equals to {C}
Q: How much is {A} plus {B}? A:	Q: How much is {A} minus {B}? A:
Q: What is {A} plus {B}? A:	Q: What is {A} minus {B}? A:
Q: What is the result of {A} plus {B}? A:	Q: What is the result of {A} minus {B}? A:
Q: What is the sum of {A} and {B}? A:	Q: What is the difference of {A} and {B}? A:
Multiplication	**Division**
{A} * {B} = {C}	{A} / {B} = {C}
{A} times {B} equals to {C}	{A} over {B} equals to {C}
The product of {A} and {B} is {C}	The ratio of {A} and {B} is {C}
The product of {A} and {B} equals to {C}	The ratio of {A} and {B} equals to {C}
The product of {A} and {B} equals to {C}	The ratio of {A} and {B} equals to {C}
Q: How much is {A} times {B}? A:	Q: How much is {A} over {B}? A:
Q: What is {A} times {B}? A:	Q: What is {A} over {B}? A:
Q: What is the result of {A} times {B}? A:	Q: What is the result of {A} over {B}? A:
Q: What is the product of {A} and {B}? A:	Q: What is the ratio of {A} and {B}? A:

Figure 8: Templates used in this work follow the formations of "Equation", "Statement", "Question-Answer".

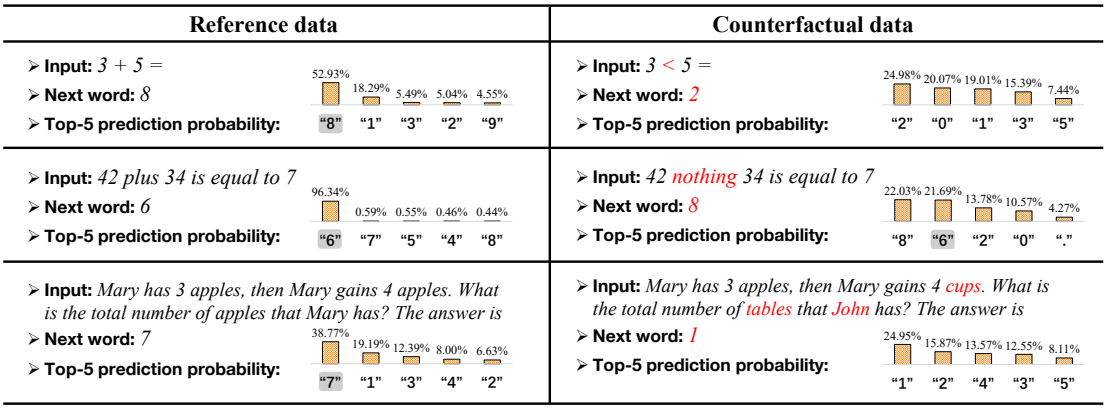


Figure 9: Examples of reference data (with addition logic) and counterfactual data (without addition logic). Given the input sentence, the results of next word prediction are provided by LLaMA2-7B.

We have included a list of 36 templates used in this work as shown in Figure 8. All these templates share the same calculation logic. we sample the <A> and <B> from $\{1, \cdots, 9\}$, since LLaMA2 tokenizes each digit individually (e.g., '42' is tokenized to '4' and '2'). Based on the above templates, we generate the sentences that the LLMs can predict the addition result {C} correctly as the reference data X_{r}. We generate the counterfactual data X_{c} following the principles depicted in Section 3, where we replace the words (*e.g.*, "plus", "minus", "times", "ratio") with a randomly-selected term from the set {"none", "nothing", $\cdots$, "null"}, and replace the operations (*e.g.*, "+", "-", "*", "/") with a randomly-selected term from the set {"<", ">", $\cdots$, "@"}. We show three cases in Figure 9 with the inspection into the top-5 prediction probability of LLaMA2-7B. Moreover, in Figure 10, we also construct several different types of linguistic meanings for the addition task: "time span" and "object accumulation". For the templates 1-8 of "time span", we sample from a

curated list of common words. For example, we select <EVENT> from {"war", "conflict", ··· , "project"}[4], <VERB> from {"last", "span", ··· , "extend"}, <MONTH> from {"Jan.", "Feb.", ··· , "Dec."}, and <YYY> from {100, ··· , 199}. For the templates 9-12 of "object accumulation", we sample <OBJECT> from {"apple", "orange", ··· , "pear"}, <VERB> from {"get", "obtain", ··· , "acquire"}, and each <NAME> was randomly selected from a pool of 100 English first names.

1. The <EVENT> <VERB> {A} years from the year <YYY>{B} to the year <YYY>{C}
2. The <EVENT> <VERB> {A} years from <YYY>{B} to <YYY>{C}
3. The <EVENT> <VERB> {A} days from <MONTH> {B} to <MONTH> {C}
4. The <EVENT> will <VERB> {A} days from <MONTH> {B} to <MONTH> {C}
5. The <EVENT> <VERB> {A} hours from {B} pm to {C}
6. The <EVENT> will <VERB> {A} hours from {B} pm to {C}
7. The <EVENT> <VERB> {A} hours from {B} am to {C}
8. The <EVENT> will <VERB> {A} hours from {B} am to {C}
9. <NAME> has {A} <OBJECT>, then <NAME> <VERB> {B} <OBJECT>. What's the total number of <OBJECT> that <NAME> has? The answer is {C}
10. <NAME> <VERB> {A} <OBJECT>, and <NAME2> <VERB> {B} <OBJECT>. What's the total number of <OBJECT> that they <VERB>? The answer is {C}
11. <NAME> has {A} <OBJECT>, and <NAME2> has {B} <OBJECT>. What's the total number of <OBJECT> that they have? The answer is {C}
12. <NAME> <VERB> {A} <OBJECT> yesterday, and <NAME> <VERB> {B} <OBJECT> today. What's the total number of <OBJECT> that <NAME> <VERB>? The answer is {C}

Figure 10: Additional templates used in the addition task, involve different linguistic meanings like "time span" (1-8) and "object accumulation" (9-12).

B. Evaluate the Effect of Attention Heads.

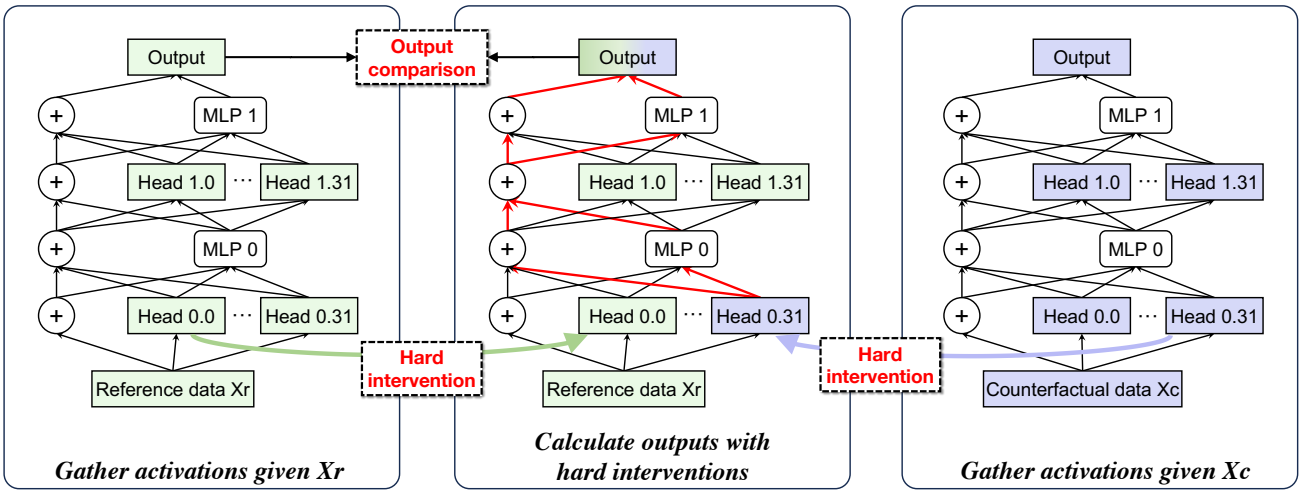


Figure 11: A case illustration of the method "path patching". It measures the importance of forward paths (*i.e.*, the red lines that originate from Head 0.31 to Output) for the two-layer transformer in completing the task on reference data.

Path Patching. To discover the cause of the predicted answer, we employ the causal intervention technique known as *path patching* (Goldowsky-Dill et al., 2023; Wang et al., 2023a). This approach is highly effective in analyzing the causal relationship between two computation nodes (Sender → Receiver). This helps us determine whether Sender is the cause of Receiver, and the connections between them are important for the model in implementing the task.

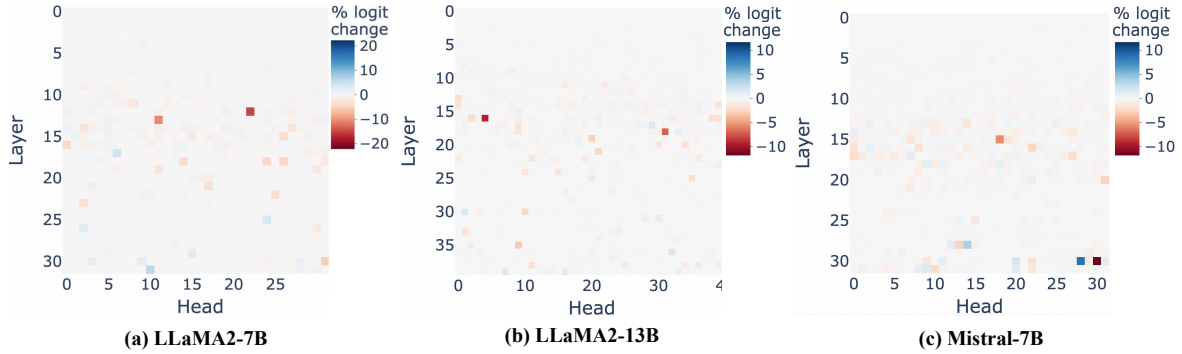
Specifically, the entire process of path patching is shown in Figure 11, where the node pair Sender → Receiver is set as Head

[4]We empirically find that the specific choice of words does not affect the results, as long as they meet similar semantics.

$0.31 \rightarrow$ Output. Firstly, given reference data X_{r} and counterfactual data X_{c}, the activations of all heads are gathered for preparation of the later perturbation. Then, we do a hard intervention on the Head 0.31 that is perturbated to its activation on X_{c}, where the effect will be further propagated to the Ouput node along with a set of paths $\mathcal{P}$. To ensure an independent observation of the impact from the Head 0.31, $\mathcal{P}$ comprises the forward pathways through residual connections and MLPs except for the other attention heads (*e.g.*, Head $0.0, \cdots, 0.30, 1.0, \cdots, 1.31$). Thus we do a hard intervention on the other heads by freezing their activations on X_{r}. Finally, we obtain the final output logits to measure the impact of this perturbation. If there is a significant change in final logits, then the patched paths: Sender $\rightarrow$ Receiver are essential for the model in completing the task.

In this work, to identify the important heads contributing to the calculation task, we scan through all heads as the Sender node denoted by h, and set the Receiver node as output $logits$, and measure the changes in the output logit of ground-truth token $\{\mathtt{C}\}$. Pathways $h \rightarrow logits$ that are critical to the model's computation should induce a large drop in the logit of token $\{\mathtt{C}\}$ after patching. Notably, since the residual operations and MLPs compute each token separately (Elhage et al., 2021), patching the head output at the END position (*i.e.*, the position of the last token in the input sentence) is enough to measure the effects on the next token prediction.

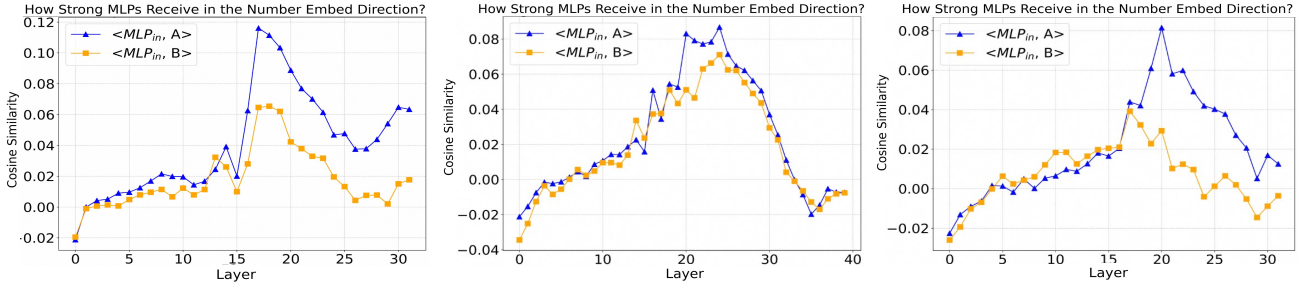
C. More Results of Other LLMs.



(a) LLaMA2-7B (b) LLaMA2-13B (c) Mistral-7B

Figure 12: Comparison of the results of path patching experiments on LLaMA2-7B, LLaMA2-13B, and Mistral-7B (Jiang et al., 2023) across four mathematical tasks. For each head/MLP, a darker color indicates a larger logit difference from the original model before patching.

Key Component Identification. In Figure 12, we further report the results of key components identification of other models (*e.g.*, LLaMA2-13B and Mistral-7B). For example, LLaMA2-13B comprises 40 layers and 40 attention heads per attention layer. The three models of different size exhibit similar phenomena that the calculation-related key heads (*e.g.*, 16.4, 18.31) are distributed sparsely in the middle layers.



(a) LLaMA2-7B (b) LLaMA2-13B (c) Mistral-7B

Figure 13: We investigate the projection of each MLP layer input (MLP_{in}) along the direction of number token $\{\mathtt{A}\}$, $\{\mathtt{B}\}$, respectively.

Key MLPs Behavior. In Figure 13, the similarities of MLP input and number operands $\{\mathtt{A}\}$/$\{\mathtt{B}\}$ across all models demonstrate ascending and descending trends. Specifically, the pivotal points for these trends, delineated as (*start-*